SeqTrack: Sequence to Sequence Learning for Visual Object Tracking

Chen et al.

CVPR 2023

인공지능융합학과
이경민

Motion Tracking - 강의 내용

Optical Flow	픽셀의 움직임 추정
Feature Tracking	특징점을 여러 frame에서 추적
Visual Object Tracking	특정 target의 위치를 계속 추정 ↑ SeqTrack이 다루는 문제

강의에서 다룬 Visual Tracking과 같은 맥락의 문제를, SeqTrack은 Transformer 기반으로 새롭게 정의했다.

기존 Tracking 방식의 문제

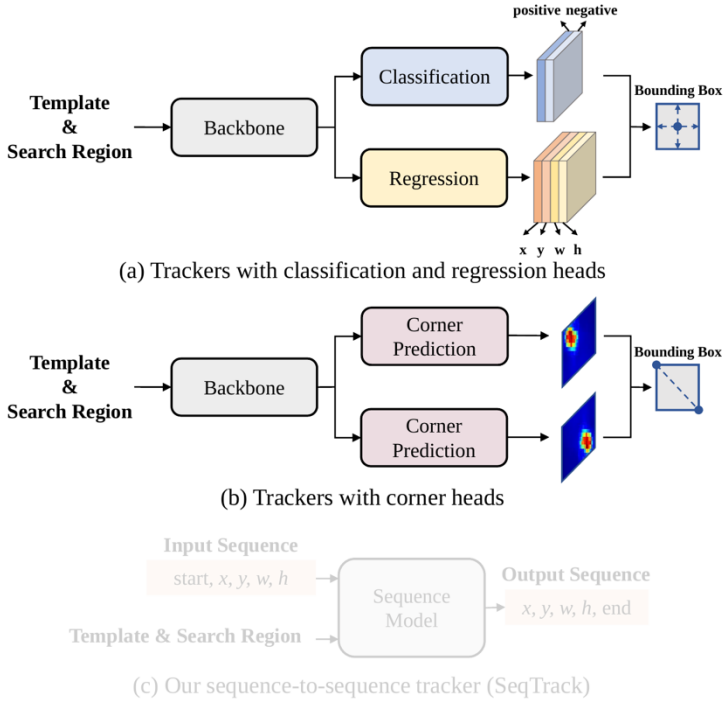


Figure 1, Chen et al.

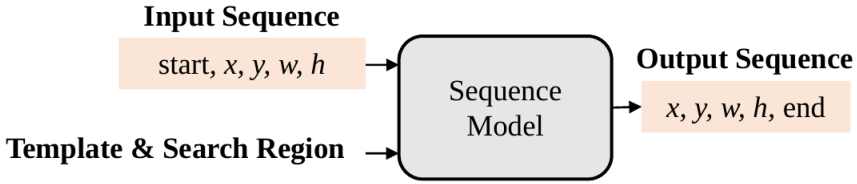
(a) Classification + Regression

(b) Corner Prediction

- task-specific head, loss가 여러 개 필요하다

Tracking을 더 단순하게 만들 수 없을까?

SeqTrack의 핵심 아이디어



(c) Our sequence-to-sequence tracker (SeqTrack)

Figure 1(c), Chen et al.

Bounding box를 예측하는 문제 → **Bounding box를 sequence로 생성하는 문제**

- Bounding box → discrete tokens (vocabulary 4,000)
- [x → y → w → h]
- Autoregressive generation

"위치를 한 번에 회귀하는 것이 아니라, 좌표를 문장처럼 하나씩 생성한다."

SeqTrack Architecture

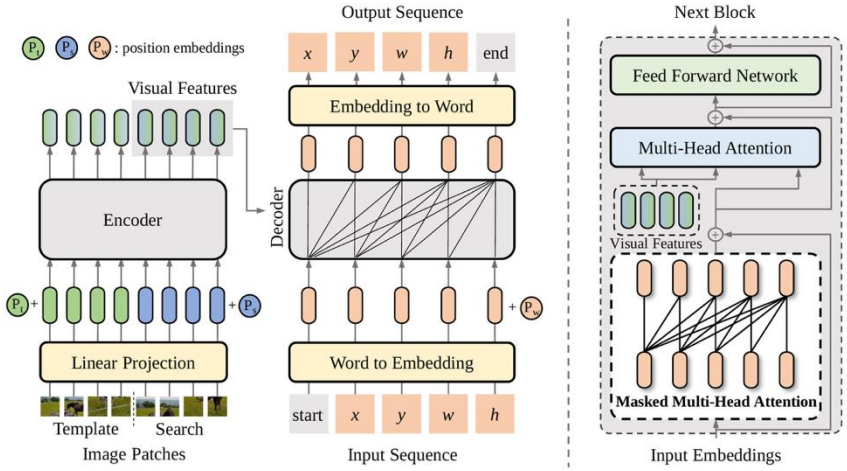


Figure 2, Chen et al.

Encoder

template과 search image를 patch로 나누고,

ViT가 visual feature를 추출

Decoder

Causal Transformer. encoder feature를 참고해

x → y → w → h 순서로 token 생성

Training & Inference

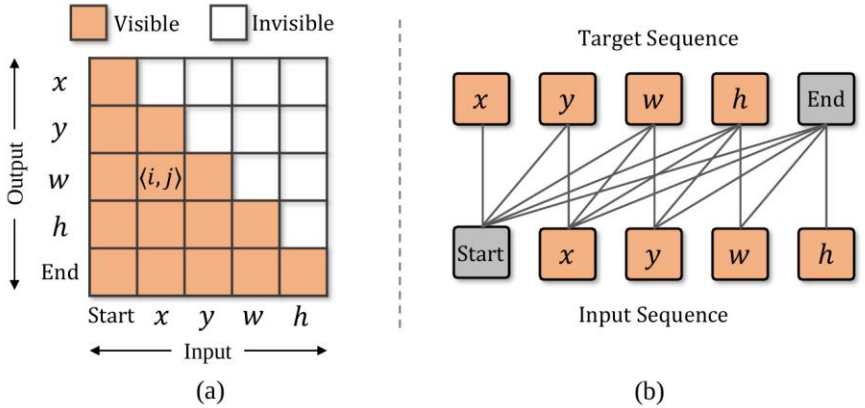


Figure 3, Chen et al.

Training

- Input: [start, x, y, w, h]
- Target: [x, y, w, h, end]
- 이전 token으로 다음 token 예측
- Cross-entropy loss

Inference

- start만 입력해서 x 생성
- [start, x]로 y 생성
- 같은 방식으로 w, h까지 반복

START → x → y → w → h

Results: Table 2

	Method	LaSOT [16]			LaSOT$_{ext}$ [15]			TrackingNet [37]			GOT-10k* [20]		
		AUC	P_{Norm}	P	AUC	P_{Norm}	P	AUC	P_{Norm}	P	AO	$SR_{0.5}$	$SR_{0.75}$
Seq2Seq	SeqTrack-L384	**72.5**	81.5	**79.3**	**50.7**	**61.6**	57.5	**85.5**	**89.8**	**85.8**	**74.8**	81.9	**72.2**
	SeqTrack-L256	72.1	**81.7**	79.0	50.5	61.5	57.2	85.0	89.5	84.9	74.5	83.2	72.0
	SeqTrack-B384	71.5	81.1	77.8	50.5	**61.6**	**57.5**	83.9	88.8	83.6	74.5	84.3	71.4
	SeqTrack-B256	69.9	79.7	76.3	49.5	60.8	56.3	83.3	88.3	82.2	74.7	**84.7**	71.8
Corner Prediction	SimTrack [4]	70.5	79.7	-	-	-	-	83.4	87.4	-	69.8	78.8	66.0
	Mixformer-L [10]	70.1	79.9	76.3	-	-	-	83.9	88.9	83.1	-	-	-
	Mixformer-22k [10]	69.2	78.7	74.7	-	-	-	83.1	88.1	81.6	70.7	80.0	67.8
	AiATrack [17]	69.0	79.4	73.8	47.7	55.6	55.4	82.7	87.8	80.4	69.6	63.2	80.0
	UTT [33]	64.6	-	67.2	-	-	-	79.7	-	77.0	67.2	76.3	60.5
	CSWinTT [44]	66.2	75.2	70.9	-	-	-	81.9	86.7	79.5	69.4	78.9	65.4
	STARK [53]	67.1	77.0	-	-	-	-	82.0	86.9	-	68.8	78.1	64.1
Classification + Regression	OSTrack-384 [55]	71.1	81.1	77.6	50.5	61.3	**57.6**	83.9	88.5	83.2	73.7	83.2	70.8
	OSTrack-256 [55]	69.1	78.7	75.2	47.4	57.3	53.3	83.1	87.8	82.0	71.0	80.4	68.2
	SwinTrack [29]	71.3	-	76.5	49.1	-	55.6	84.0	-	82.8	72.4	-	67.8
	RTS [39]	69.7	76.2	73.7	-	-	-	81.6	86.0	79.4	-	-	-
	Unicorn [52]	68.5	-	-	-	-	-	83.0	86.4	82.2	-	-	-
	SLT [24]	66.8	75.5	-	-	-	-	82.8	87.5	81.4	67.5	76.5	60.3
	SBT [50]	66.7	-	71.1	-	-	-	-	-	-	70.4	80.8	64.7
	ToMP [34]	68.5	79.2	73.5	45.9	-	-	81.5	86.4	78.9	-	-	-
	KeepTrack [35]	67.1	77.2	70.2	48.2	-	-	-	-	-	-	-	-
	AutoMatch [57]	58.3	-	59.9	-	-	-	76.0	-	72.6	65.2	76.6	54.3
	TransT [7]	64.9	73.8	69.0	-	-	-	81.4	86.7	80.3	67.1	76.8	60.9
	TrDiMP [48]	63.9	-	61.4	-	-	-	78.4	83.3	73.1	68.8	80.5	59.7
	SiamAttn [56]	56.0	64.8	-	-	-	-	75.2	81.7	-	-	-	-
	SiamBAN [8]	51.4	59.8	-	-	-	-	-	-	-	-	-	-
	DSTrpn [42]	43.4	54.4	-	-	-	-	64.9	-	58.9	-	-	-
	Ocean [58]	56.0	65.1	56.6	-	-	-	-	-	-	61.1	72.1	47.3
	SiamR-CNN [47]	64.8	72.2	-	-	-	-	81.2	85.4	80.0	64.9	72.8	59.7
	DiMP [3]	56.9	65.0	56.7	39.2	47.6	45.1	74.0	80.1	68.7	61.1	71.7	49.2
	SiamPRN++ [26]	49.6	56.9	49.1	34.0	41.6	39.6	73.3	80.0	69.4	51.7	61.6	32.5
	ATOM [12]	51.5	57.6	50.5	37.6	45.9	43.0	70.3	77.1	64.8	55.6	63.4	40.2
	MDNet [38]	39.7	46.0	37.3	27.9	34.9	31.8	60.6	70.5	56.5	29.9	30.3	9.9

Table 2, Chen et al.

SeqTrack-L384

LaSOT AUC **72.5**

TrackingNet AUC **85.5**

GOT-10k AO **74.8**

Ablation Study: Table 4

#	Method	LaSOT	GOT-10k	Δ
1	Baseline	69.2	73.7	–
2	Autoregressive → Bidirectional	64.8	72.4	**-2.9**
3	Joint → Separate	62.0	66.1	**-7.4**
4	$[x,y,w,h] \rightarrow [w,h,x,y]$	67.1	71.8	**-2.0**
5	$[x,y,w,h] \rightarrow [x_{min},y_{min},x_{max},y_{max}]$	68.3	70.7	**-2.0**
6	Concat of Search-Template	69.6	73.3	**-0.0**
7	Avg. of Search-Template	69.2	72.2	**-0.8**
8	+ Likelihood-based Online Update	69.9	76.1	**+1.6**
9	+ Naive Online Update	69.3	73.1	**-0.3**
10	− Window Penalty	68.8	73.1	**-0.5**

Table 4, Chen et al.

Autoregressive → Bidirectional

-2.9 autoregressive generation이 중요

Joint → Separate (Encoder)

-7.4 template과 search를 함께 encoding하는 게 중요

Likelihood-based Online Update

+1.6 confidence를 이용한 template update가 효과적

Contributions

1. New Formulation

Tracking을 sequence generation problem으로 재정의

2. Simple Framework

복잡한 classification / regression / corner head 제거

3. Competitive Performance

단순한 구조인데도 SOTA 수준의 성능

Limitations

- Explicit re-detection module이 없다
- Target이 화면 밖으로 나가거나 occlusion, distractor가 있는 상황에 약하다
- Temporal modeling이 제한적이다
- Future work: 전체 video sequence 활용

감사합니다.

Paper: CVPR 2023 Open Access
Code: github.com/microsoft/VideoX/SeqTrack
https://ieeexplore.ieee.org/document/10203645